

SHIVAJI UNIVERSITY, KOLHAPUR



Dr. D. Y. Patil Pratishthan's College of Engineering

Salokhenagar, Kolhapur

2024-2025

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(DATA SCIENCE)



A
PROJECT REPORT
ON

“AI-ENHANCED LEARNING ASSISTANT”

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CERTIFICATE

Certified that the Project topic entitled “AI-ENHANCED LEARNING ASSISTANT” a bonafide work carried out by Siddhesh Agre, Vaishnavi Dawar, Bajarang Dhamanekar, Pavan Jadhav, Sakshi Jadhav in partial fulfilment for the award of Degree of Bachelor of Technology in 8th Semester of the SHIVAJI UNIVERSITY, KOLHAPUR during the year 2024-2025. It is certified that all corrections/ suggestions indicated for Internal Assessment have been incorporated in the report deposited in the Department Library. The Project report has been approved as it satisfies the Academic requirement in respect of Project work prescribed for BACHELOR OF TECHNOLOGY DEGREE.

Prof. Rohit S. Raut
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EXAMINERS

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- 1.
- 2.

DECLARATION

We, the undersigned, students of B.Tech. Computer Science and Engineering (Data Science) declare that the project work report entitled “**AI ENHANCED LEARNING ASSISTANT**” written and submitted under the guidance of **Prof. Rohit S. Raut**. The empirical findings in this report are based on the data collected by us. The matter assimilated in this report is not reproduction from any readymade report.

Place: Salokhenagar, Kolhapur

Date:

Yours Sincerely,

Sign

Mr. Siddhesh S. Agre

Miss. Vaishnavi R. Dawar

Mr. Bajarang K. Dhamanekar

Mr. Pavan D. Jadhav

Miss. Sakshi H. Jadhav

ACKNOWLEDGEMENT

*The sense of contentment and relation that accompanies the successful completion of the project “**AI-ENHANCED LEARNING ASSISTANT**” would be incomplete without mentioning the names of those people who helped us in accomplishing the project. Those people whose Constant guidance and encouragement resulted in its realization.*

*We take this opportunity to thank our Campus Director **Dr. A. M. Mane** for providing a constant support and resources that helped us in completing the task.*

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*We express a deep sense of gratitude to our **H.O.D Prof. R. S. Raut** for providing the inspiration required for taking the Project to its completion.*

*We convey our heart full thanks to our guide **Prof. R. S. Raut** for the guidance and inspiration they had given during the course of completion of our project. We also thank all the staff member, teaching and non-teaching for helping us to accomplish this project.*

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ABSTRACT

The AI-enhanced Question Answer Generation and Evaluation System is an educational assessment method providing answer generation, evaluation and scoring through Natural Language Processing (NLP) and Machine Learning (ML) as a core tool. In this paper, we describe the main aspects of this AI system, which analyzes a given text to extract pertinent answer questions and then grade students' answers on their matching with predefined answer keys. Natural language processing methods are used for concept extraction to achieve an equitable assessment model. Learning objectives and targeted question-writing strategies are applied in order to generate multiple question-writing strategies for different levels of students. The answering evaluation module uses ML models to assess student responses against the provided answer, analyzing analogy-based error detection, semantic similarity and accuracy-scoring to measure relevance and accuracy. Student errors are explained with the specific reasons within each error.

Students benefit from these comments by being able to overcome subsequent attempts. Major components of assessment involve preprocessing, scoring, semantic analysis, score feedback and automated creation of scorecards. With these AI features integrated, consistency and objectivity of time allocation, personalized feedback, adjusted demand, and overall assessment can be improved while still maintaining scalability and credibility.

CHAPTER 1
INTRODUCTION

1.1 Introduction

The AI-Enhanced Question Answer Generation and Evaluation System represents a cutting-edge advancement in educational technology, harnessing the power of advanced natural language processing (NLP) to revolutionize the way we create and assess questions and answers. This sophisticated system is designed to automatically generate a wide range of contextually relevant questions from textual content, offering a significant enhancement to traditional evaluation methods. By leveraging state-of-the-art NLP techniques, the system ensures that the questions produced are not only diverse but also tailored to the specific content and context, enabling a more comprehensive and nuanced assessment of understanding.

Central to this system is its evaluation component, which utilizes machine learning algorithms to rigorously assess the accuracy and quality of the answers provided. This automated evaluation process offers detailed feedback and scoring, allowing for a more objective and scalable approach to assessment. The machine learning models embedded in the system are capable of analyzing responses with a high degree of precision, providing valuable insights into areas where learners may need further improvement.

The integration of such advanced technology into educational and training processes brings numerous benefits. It ensures consistency in the assessment process, as the system applies uniform criteria to evaluate responses, reducing potential biases associated with human graders. Additionally, it enhances efficiency by automating the generation and assessment of questions, thereby saving time and resources while increasing the scalability of the evaluation process. This means that educators and trainers can focus more on personalized instruction and support, while the system handles the repetitive and time-consuming aspects of question generation and answer evaluation.

In summary, the AI-Enhanced Question Answer Generation and Evaluation System marks a significant leap forward in the field of educational assessment. By combining innovative question generation capabilities with robust automated evaluation, it offers a powerful tool for improving the quality and effectiveness of educational and training programs. This approach not only ensures more accurate and insightful assessments but also supports a more engaging and personalized learning experience.

1.2 Need of Project

- Scalability Issues: Traditional assessment methods struggle to handle large numbers of learners effectively.
- Time-Consuming Processes: Manual question creation and answer evaluation require significant time and effort from educators.
- Human Error and Bias: Evaluations done by humans can be inconsistent and subjective, affecting fairness and accuracy.
- Demand for Personalization: Modern education emphasizes personalized learning, which is difficult to achieve with manual methods.
- Lack of Efficiency: Manual assessments are not resource-efficient and can slow down the learning process.
- Need for Objective Evaluation: Automated systems can apply uniform criteria for fair and unbiased assessments.
- Growing Online and Remote Learning: The rise of e-learning and remote education calls for scalable and intelligent assessment tools.
- Real-Time Feedback Requirement: Learners benefit from immediate, detailed feedback, which manual grading cannot consistently provide.
- Enhanced Learning Outcomes: Automatically generated, context-aware questions help in deeper understanding and better retention.
- Support for Educators: Automation allows educators to focus more on instruction and learner engagement rather than repetitive evaluation tasks

1.3 Methodology

1. Technology Stack

Backend Framework: Flask (Python)

Frontend: HTML, CSS, JavaScript

Database: MySQL

ML/NLP Libraries: nltk, scikit-learn

File Processing: pdfkit, werkzeug, pickle

Deployment Tools: Flask, WSGI

2. System Architecture

The system is built in modular fashion:

Authentication Module – Sign-up, log in, log out, forgotten password, and updating profile.

Test Generation Module – employs NLP to create objective or subjective questions.

Evaluation Module – Compares student responses to reference responses based on cosine similarity.

PDF Report Generator - Provides the functionality to export test results to PDF.

3. User Profile and Authentication Management

They may register with information like last name, first name, password, and email. They are managed by Flask's session system. User data is securely saved in MySQL database. Profile picture upload is handled by file storage and images are served using the Flask static route.

4. Question Generation System

Objective Question Generation: Utilizes the ObjectiveTest class (objective.py) to create multiple-choice questions.

Subjective Question Generation: Uses the SubjectiveTest class (subjective.py) to create explanatory questions. TF-IDF and nltk NLP Preprocessing are used in keyword extraction and text analysis. The user selects the test type and question number, and it is dynamically processed.

5. Answer Evaluation System

The respondents write out their responses in.txt form. A pre-loaded reference answer is read from a static file

(reference_answers.txt). TF-IDF Vectorization and Cosine Similarity (borrowed from scikit-learn) are used to compute the similarity score of the student's response to the reference response.

Score is standardized and is graded on a scale of 10.

6. PDF Report Generation

HTML output generated through rendering is sent to pdfkit via wkhtmltopdf to convert. The produced PDF file consists of the reference and the marked student response. PDF is downloaded through Flask's send_file() function.

7. Models and Data Upload Integration

The structured data or text can be uploaded by users from a server-side uploads directory. A pre-trained ML model (AI_model.pkl) is loaded using pickle when training starts. The structure facilitates retraining or reloading models for future improvements.

8. Database Design

A single table users in MySQL stores all user-related information, including image paths. The schema possesses properties like id, first_name, last_name, username, email, password.

9. Session and Security Handling of Logged-in users

Session tokens are used to manage logged-in users. Secure file handling is offered through secure_filename() in order to avoid directory traversal attacks. Passwords are plain in the demo but need to be hashed in production (using bcrypt or werkzeug.security).

10. Error Handling and Validation

Validations are performed at all steps: validation of the file type, validation of the form field, sanitization of the image filename, and check of the existence of the user.

CHAPTER 2
LITERATURE SURVEY

2. Literature Survey

Automatic question generation (AQG) is an essential task in Natural Language Processing (NLP) with significant applications in education, assessments, and AI-based tutoring systems. Several foundational works and modern research contributions have helped shape this domain. This review outlines key literature that underpins our work on both objective and subjective question generation

2.1 Evaluation Metrics for Automatic Text Generation [1]

Akhilesh Dhamdhere (2020) presents a comprehensive evaluation of metrics commonly used in automatic text generation tasks. The paper compares metrics like BLEU, ROUGE, and embedding-based scores, shedding light on their respective strengths and limitations. BLEU, for instance, while effective for n-gram overlap, often fails to capture semantic similarity, which is crucial in AQG tasks. ROUGE emphasizes recall over precision, suitable for summarization but can be misleading when applied to question generation. Embedding-based metrics (e.g., cosine similarity over sentence embeddings) offer deeper semantic comparisons and are recommended for subjective question evaluation. This work is highly relevant for your project, particularly in evaluating the quality of generated questions and ensuring alignment with human-like outputs.

2.2 Contextual Language Models and Their Role in AQG [2]

The work by Jacob Devlin et al. (2018) introduced BERT (Bidirectional Encoder Representations from Transformers), revolutionizing NLP by enabling deep bidirectional understanding of context. BERT is foundational to modern NLP systems and is particularly effective in generating subjective-type questions where deep understanding of sentence structure and semantics is critical. Your project can benefit from BERT's pre-trained models for tasks like identifying key concepts, semantic roles, and transforming declarative sentences into interrogative ones. This paper establishes the effectiveness of transfer learning and contextual embedding in boosting AQG model performance..

2.3 NLP Foundations and Core Techniques [3]

Daniel Jurafsky and James H. Martin's textbook (2021) serves as a foundational reference for understanding NLP tasks. It provides a thorough explanation of key techniques such as tokenization, part-of-speech tagging, parsing, named entity recognition, and semantic

similarity, all of which play essential roles in building robust AQG systems. These techniques are critical in preprocessing text data, identifying important tokens for question formulation, and understanding sentence structure. The theoretical and practical insights from this work help establish the NLP pipeline for both objective and subjective question generation.

2.4 Deep Learning in NLP and AQG

[4]

Palash Goyal, Sumit Pandey, and Karan Jain (2018) focus on applying deep learning models to NLP, which is particularly useful for complex generation tasks like AQG. The book emphasizes architectures such as RNNs, LSTMs, and transformers, which have been successfully employed in question generation models. These models allow the system to learn from context, generate coherent questions, and improve over time with training. Your project can draw from this work to understand the architecture best suited for generating different types of questions and implementing training mechanisms for improved accuracy.

2.5 QA Pair Generation from Text

[5]

Choi and Choi (2018) address the challenge of automatically generating Question-Answer (QA) pairs from unstructured text. This research outlines methods to extract answers and frame appropriate questions around them, mimicking the human QA process. It aligns closely with your project's ObjectiveTest and SubjectiveTest modules, where answers must be identified and converted into meaningful, contextually relevant questions. Their use of semantic roles and dependency parsing can be integrated into your own approach to improve answer selection and question generation quality.

2.6 Natural Language Techniques for AQG and Answering

[6]

The upcoming work by Choudhury and Bhattacharyya focuses on holistic AQG systems capable of producing insightful and relevant questions using advanced NLP techniques. The authors propose a pipeline combining context detection, keyphrase extraction, semantic understanding, and question formulation. This work supports the objective of building systems that can interact intelligently with textual data to produce meaningful assessment questions. Integrating their suggested methods could enhance your system's ability to handle varied text complexity and generate questions tailored to both comprehension and knowledge evaluation.

2.7 Wang et al. (2022). "Interaction Design for AI Question Generation Systems in Education"

Wang emphasizes the importance of customizable UI elements in QG systems. For example, allowing users to choose question difficulty levels, topics, or answer formats through an intuitive interface improves satisfaction and engagement.

CHAPTER 3
PROPOSED SYSTEM

3. Proposed System

3.1 Proposed Diagram:

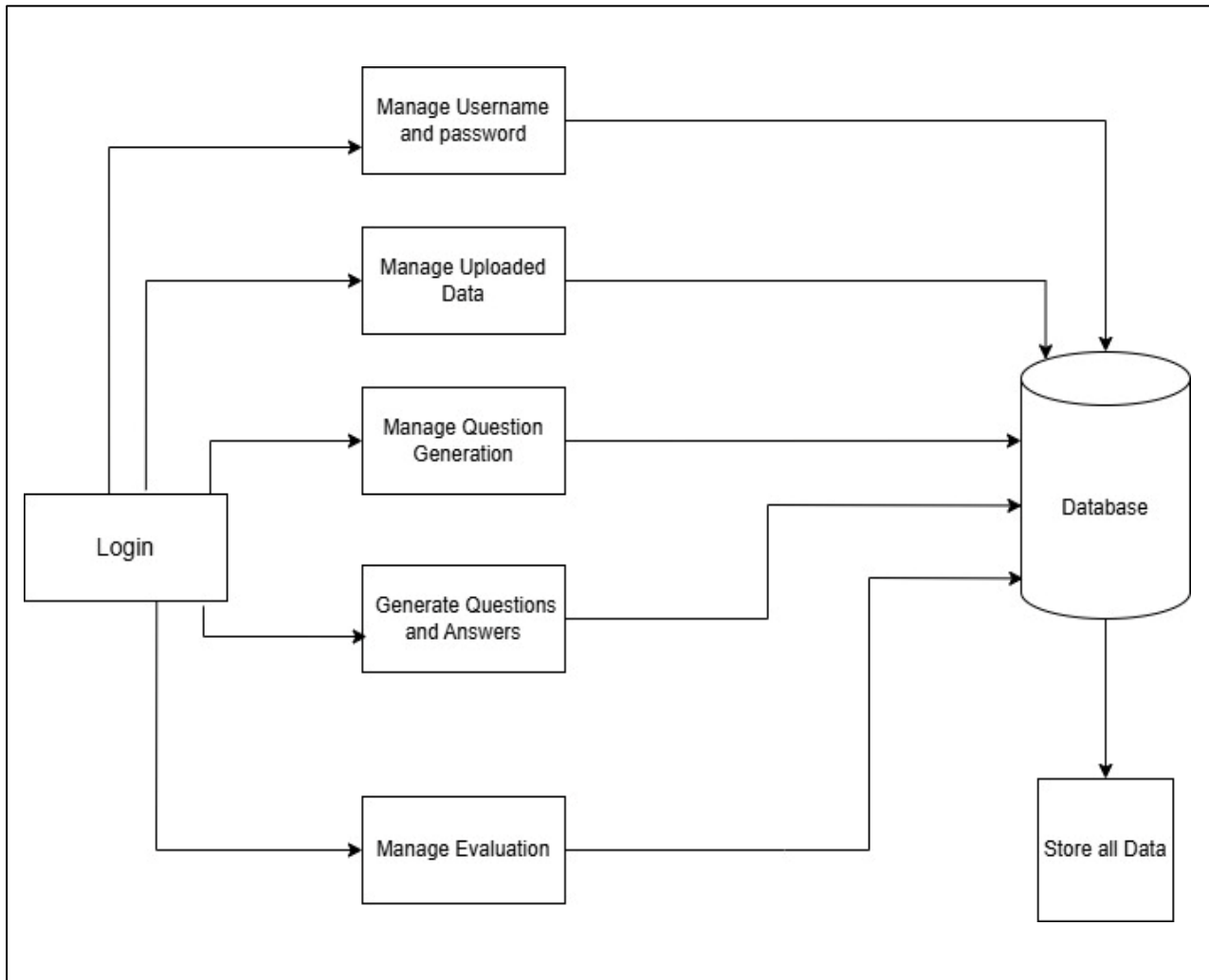


Fig. 3.1 Proposed Diagram for web Portal

3.2 Modules:

1. Data Collection Module

Sources: The data used for generating questions and evaluating answers is collected either via file uploads (e.g., PDFs, text files) or from a database (MySQL).

Data Types: Includes both textual data from files (e.g., reference_answers.txt, student answers) and image data (e.g., profile images).

Module Usage:

File Upload: upload() route handles file uploads from users, which will later be used for training models or generating questions.

2. Data Preprocessing Module

Cleaning: Involves parsing and cleaning uploaded files, such as extracting text from documents and validating file content.

Tokenization: Tokenization of textual data happens in various sections, such as when processing the student and reference answers for comparison.

Annotation: For generating tests and evaluating answers, key data points (questions, answers, scores) are extracted, which helps in further training and evaluation.

3. Model Training Module

Algorithms Used: You mentioned using ML algorithms, specifically transformers and neural networks. In the current context, a model is loaded via pickle for training and evaluation.

Training Data: The data for training models comes from uploaded files. For example, the train_model() route uses a model saved as AI_model.pkl.

Validation: A placeholder function for model training and testing is provided, though real model training and validation appear to be yet to be implemented fully.

4. Question Generation Module

Techniques: The code uses different test generation techniques based on the type of test (ObjectiveTest for MCQs and SubjectiveTest for essay-type questions).

Question Types: Supports objective and subjective question types.

Difficulty Levels: Though not fully implemented, the question_generation() route provides an interface to generate questions based on the uploaded text input.

5. Answer Evaluation Module

Automated Scoring: This module evaluates student answers against a reference answer, using cosine similarity via the `calculate_similarity()` function. It provides a score based on how closely the student's answer matches the reference answer.

6. System Integration Module

User Interface: The UI components, including forms for file uploads, login/signup, and profile management, are built using Flask's rendering capabilities. Routes like `user_dashboard()`, `update_profile()`, and `question_generation()` provide necessary UI functionality.

APIs: The system doesn't explicitly expose APIs in the traditional sense but integrates well with the frontend via Flask routes that handle dynamic HTML rendering and file management.

Scalability: The Flask app is designed to be scalable in terms of supporting different users and handling various operations like file uploads, question generation, and answer evaluation.

Module Breakdown in the Code:

Data Collection:

Routes: `upload()`, `eval_upload()`, `forgot_password()`

Data Types: Text (answers, questions), images (profile)

Data Preprocessing: Tokenization and cleaning logic are indirectly implemented through file processing and text extraction, especially in the `eval_upload()` and `calculate_similarity()` functions.

Model Training:

Route: `train_model()`

Placeholder for real model training using an AI model loaded via pickle.

Question Generation:

Route: `question_generation()`

Classes: `ObjectiveTest`, `SubjectiveTest`

Answer Evaluation:

Route: `eval_upload()`

Function: `calculate_similarity()`

System Integration: Flask routes for interaction: `home()`, `signup()`, `login()`, `update_profile()`, etc.

Integrates with MySQL to handle user information and updates.

CHAPTER 4
EXISTING SYSTEM

4.1 Existing System:

1. Early Systems: Predefined Templates and Rule-Based Algorithms

Question Generation:

Predefined Templates: Early systems for question generation relied heavily on predefined templates. These systems typically used a set of rules or templates to extract certain key phrases from a body of text and rephrase them into question formats.

Rule-Based Algorithms: These systems followed strict grammatical rules. For example, a rule might define that a sentence starting with "The capital of France is Paris" could be converted into the question "What is the capital of France?"

Automated Grading:

Keyword Matching: These systems relied on simple keyword matching techniques. They compared the keywords in the student's answer with those in the reference answer. For example, if the student's answer mentioned "Paris," it would be marked as correct for the question "What is the capital of France?".

Rule-Based Algorithms: Grading was also rule-based. The system would check whether the answer followed certain patterns, like matching specific sentence structures or phrases. If the structure matched, the answer was considered correct.

2. Modern Systems: NLP-Based Question Generation and Automated Grading

Question Generation Using NLP Techniques:

Part-of-Speech Tagging: Modern systems use Part-of-Speech (POS) tagging to identify and classify words in a sentence (e.g., nouns, verbs, adjectives). This helps in converting statements into questions. For instance, the sentence "The Eiffel Tower is in Paris" could be transformed into "Where is the Eiffel Tower?"

Dependency Parsing: This technique helps in understanding the syntactic relationships between words in a sentence. It is used to create grammatically correct questions based on the relationships between entities.

Template Generation: Some NLP systems combine templates with dynamic rule-based methods. For instance, they can generate a question based on a template like "What is the [subject] of [object]?" and fill in the blanks based on the content of the text.

Automated Grading Using NLP Techniques:

Semantic Similarity: One major advancement in grading systems is the use of semantic similarity measures, such as cosine similarity, to assess the similarity between the student's

answer and the reference answer. Instead of relying on exact keyword matches, the system evaluates how similar the meaning of the answers is. Tools like TF-IDF (Term Frequency-Inverse Document Frequency) or BERT embeddings are used for this purpose.

CHAPTER 5
SYSTEM ANALYSIS

5. System Analysis

5.1 Problem Statement

Design and develop an automated question and answer generation system, integrated with a scoring mechanism to evaluate student responses. The system should be capable of generating contextually relevant and accurate questions and corresponding answers based on input content. Additionally, it must include a robust evaluation module that compares students' answers with predefined correct answers to deliver precise and fair scoring.

The key challenges involve:

- Ensuring that the generated questions and answers are contextually appropriate, grammatically correct, and aligned with the given content.
- Implementing an effective scoring system that can accurately assess semantic similarity between student responses and the answer key.
- Providing reliable, unbiased, and consistent evaluation across a wide range of topics and question types.

5.2 System Requirements:

5.2.1 Software Requirements:

- Flask
- NLP, Deep Learning
- HTML, CSS, JavaScript,
- XAMPP
- MySQL

5.2.2 Hardware Requirements:

- Minimum 1–2 GB RAM
- Minimum 10GB Hard disk
- 2.0–2.5 GHz or 64-bit (x64) Processor

CHAPTER 6
SYSTEM DESIGN

6.1 Use Case Diagram:

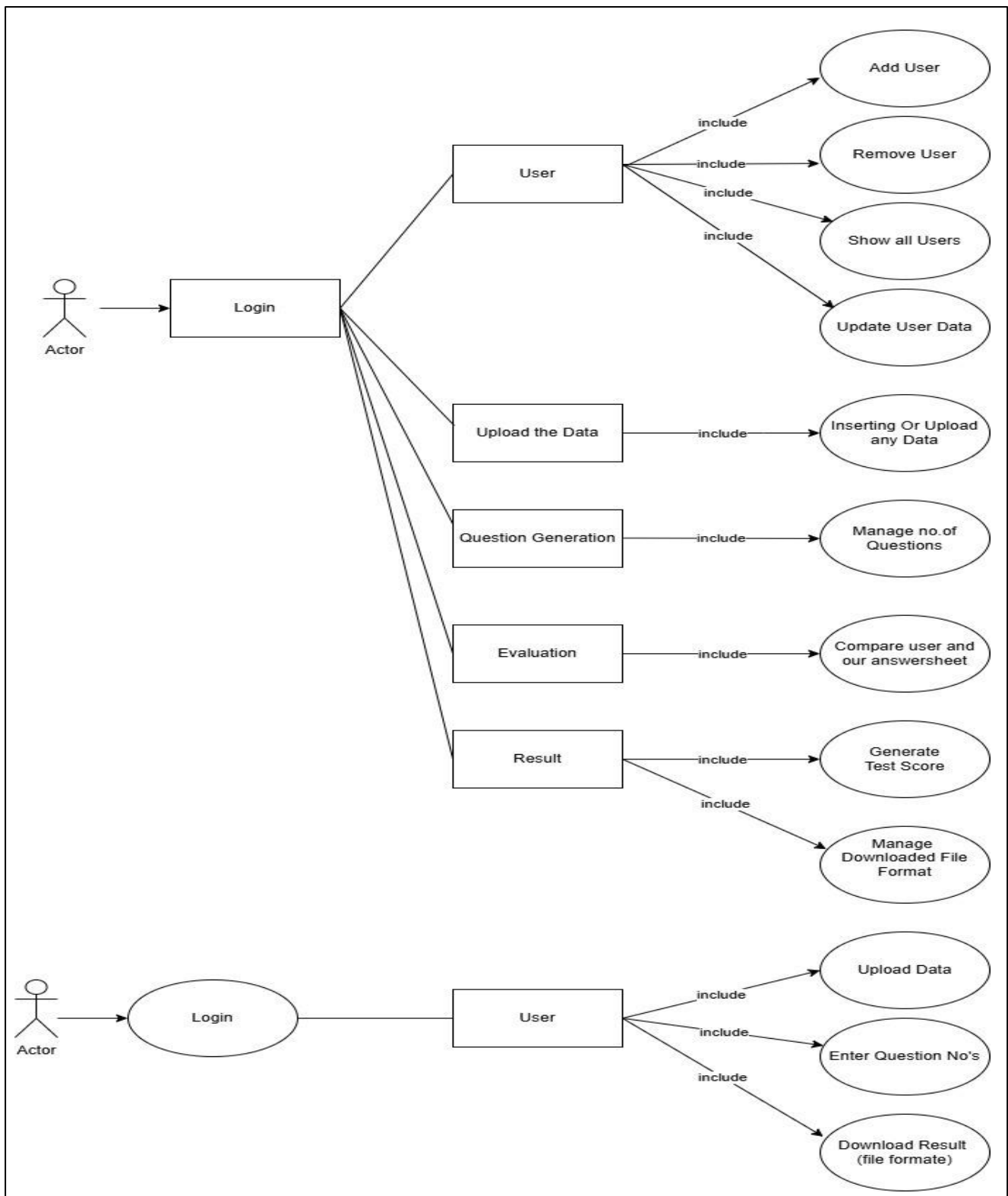


Fig. 6.1 Use Case Diagram

In above fig. 6.1, The admin can login into the system and can access through data. Admin can access the User's data, Uploaded data, questions management. Students can upload input data and download it's result also it can update his profile details.

6.2 State Diagram

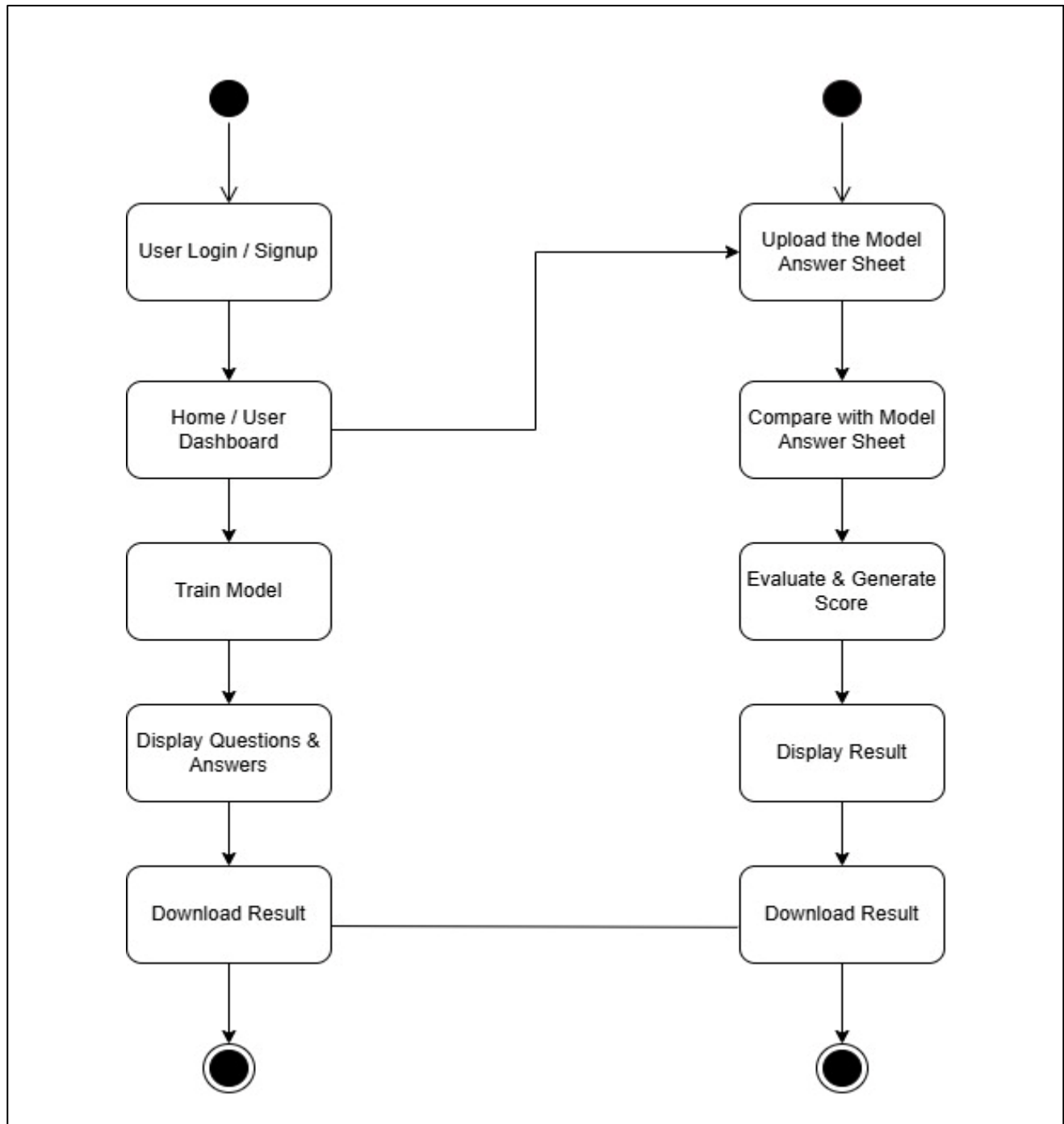


Fig 6.2 State Diagram

In above fig. 6.2, User can login successfully and go to the display dashboard and can logout after access. If the login is invalid then User are rejected to get access.

6.3 Activity Diagram

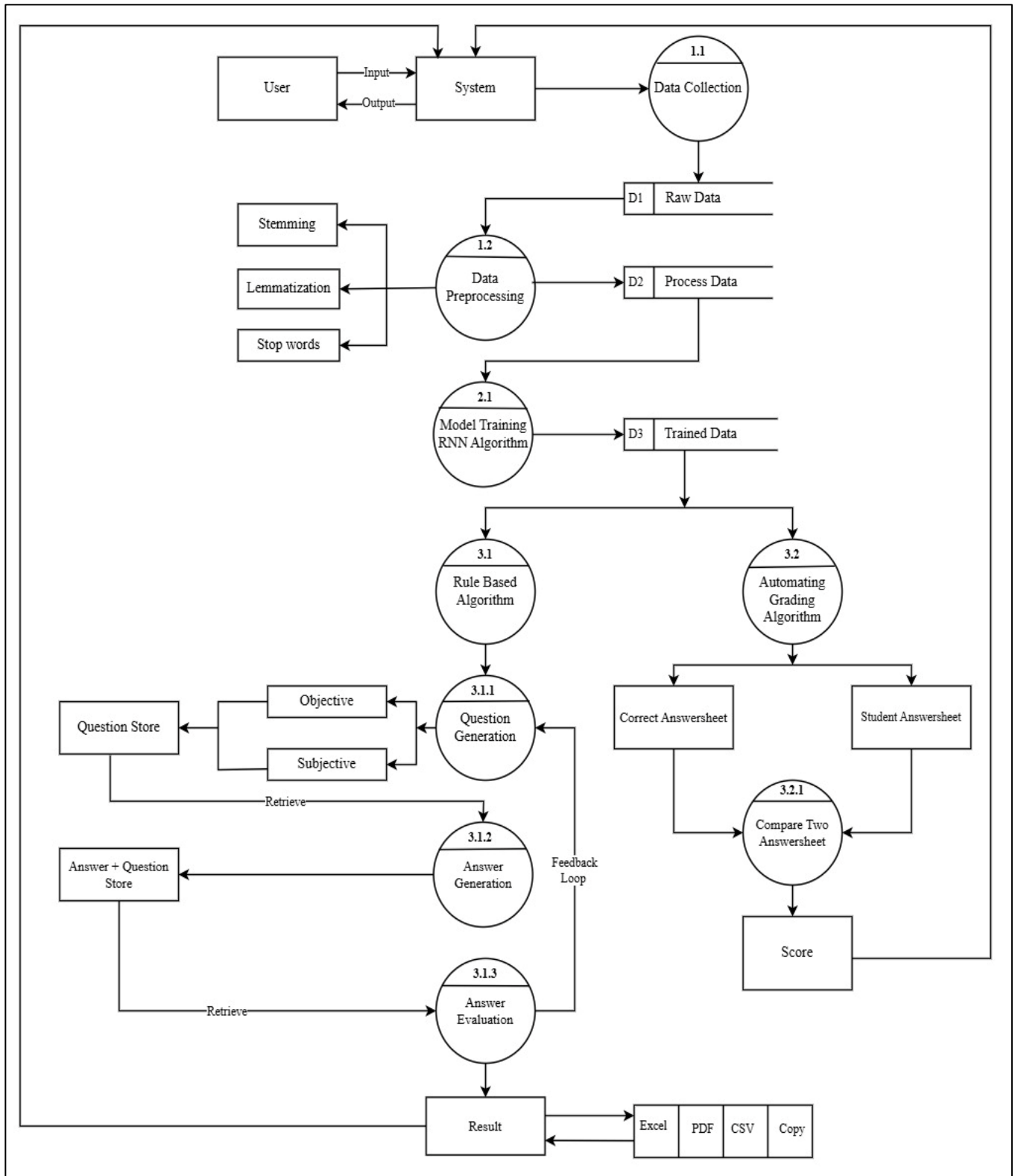


Fig. 6.3 Activity Diagram

6.4 Sequence Diagram

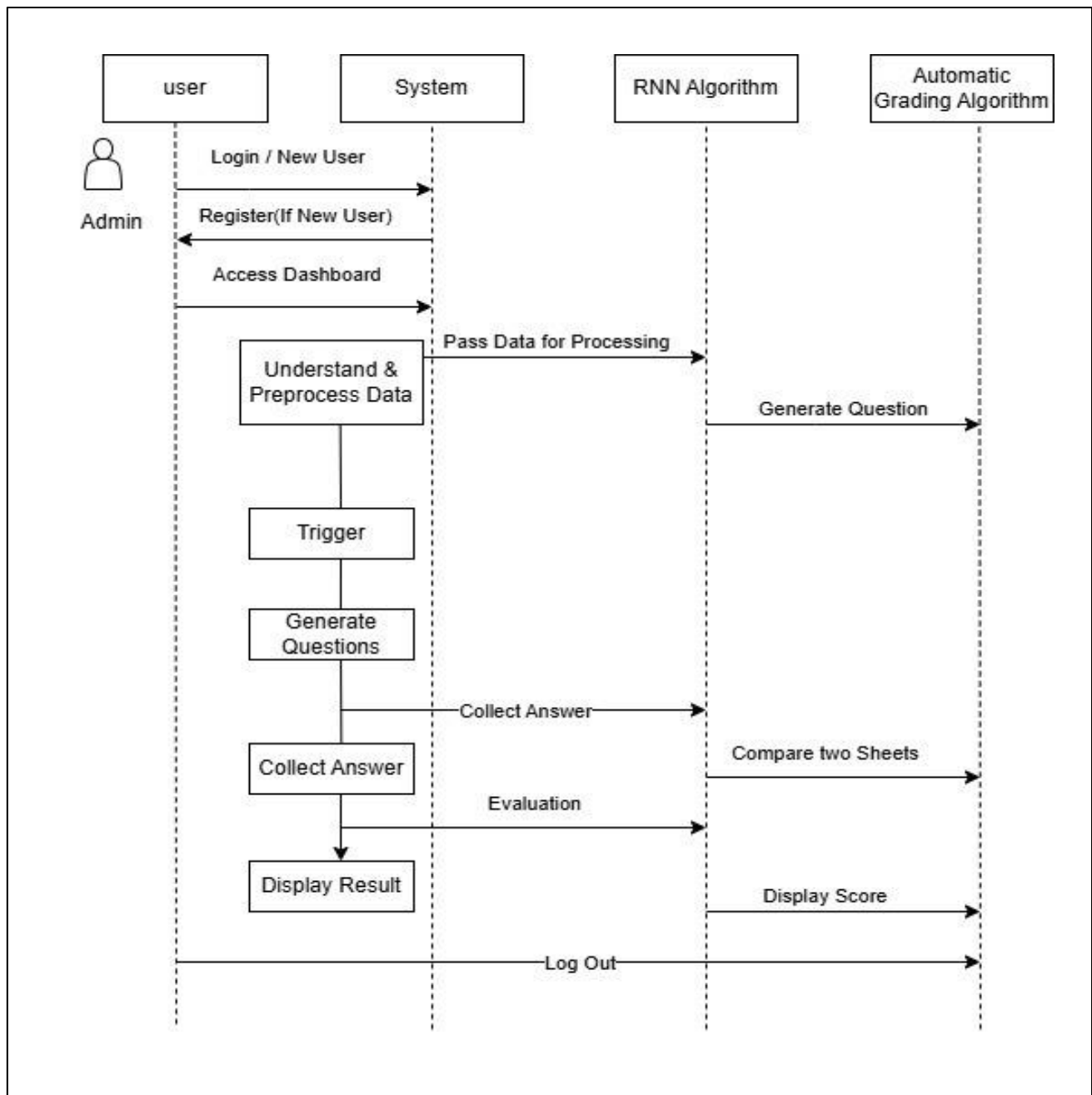


Fig. 6.4 Sequence Diagram

CHAPTER 7
SYSTEM IMPLEMENTATION

7 System Implementation

7.1 System Implementation:

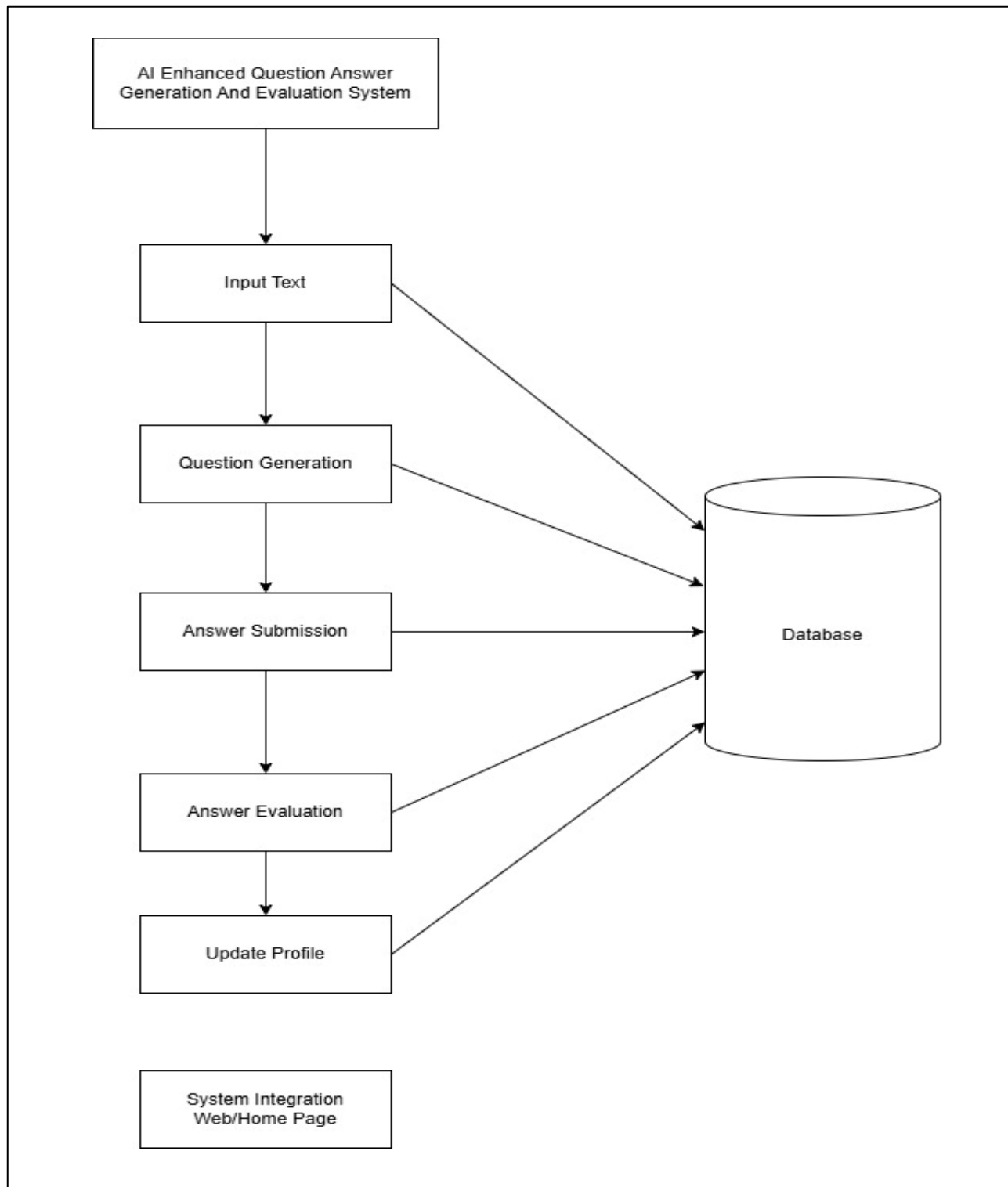
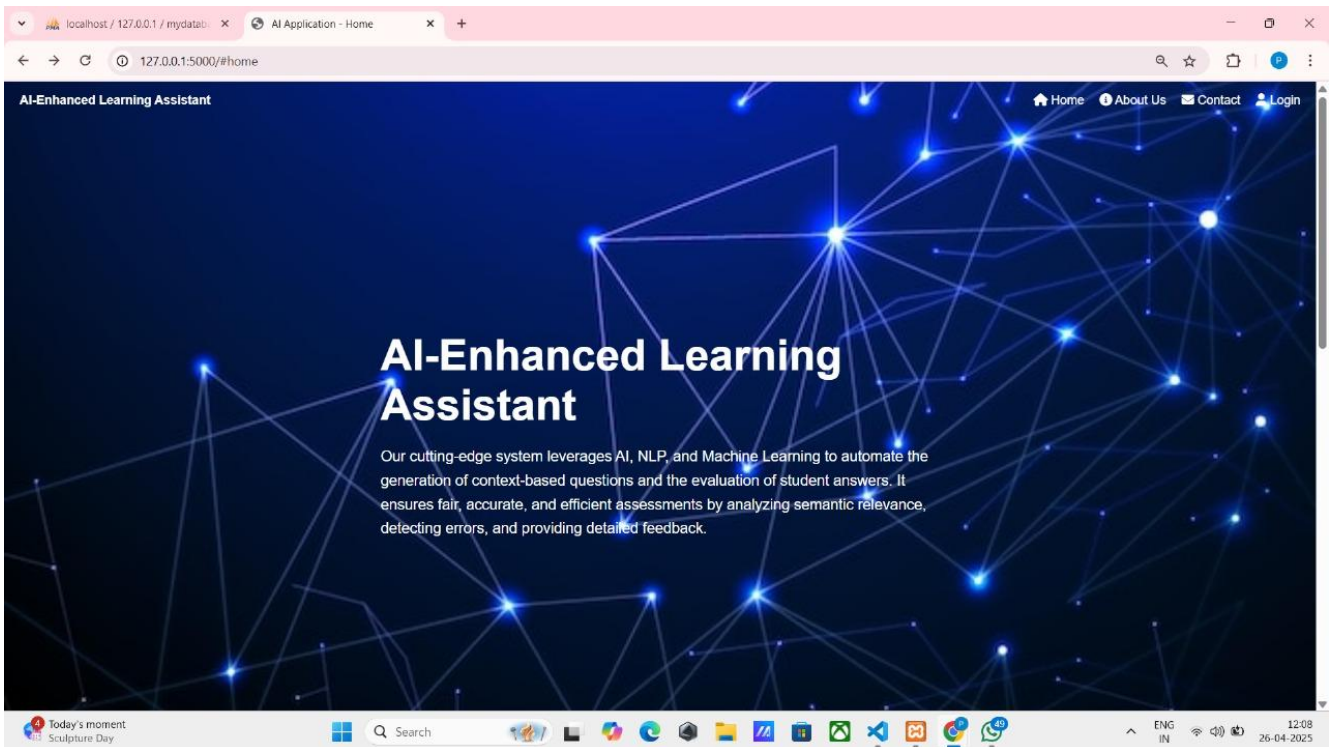


Fig. 7.1 System Implementation

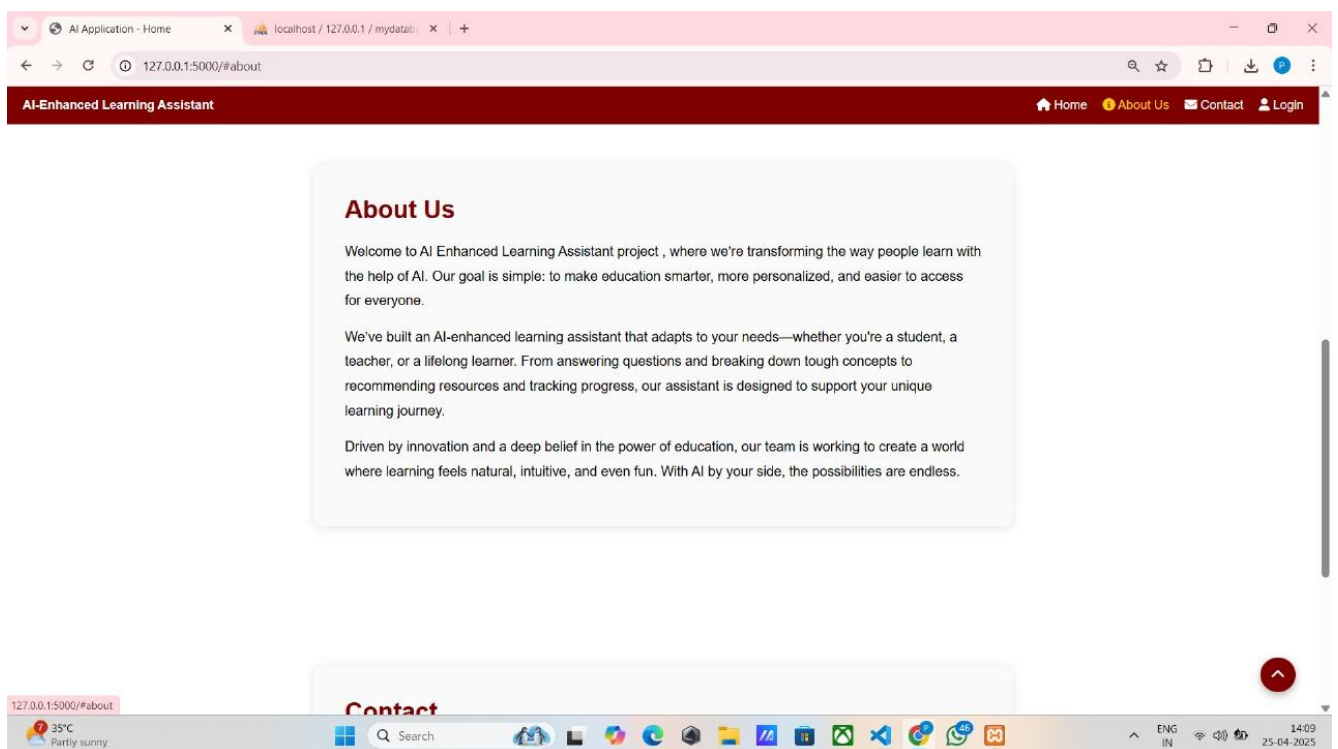
7.2 Snapshots

7.2.1 User Login

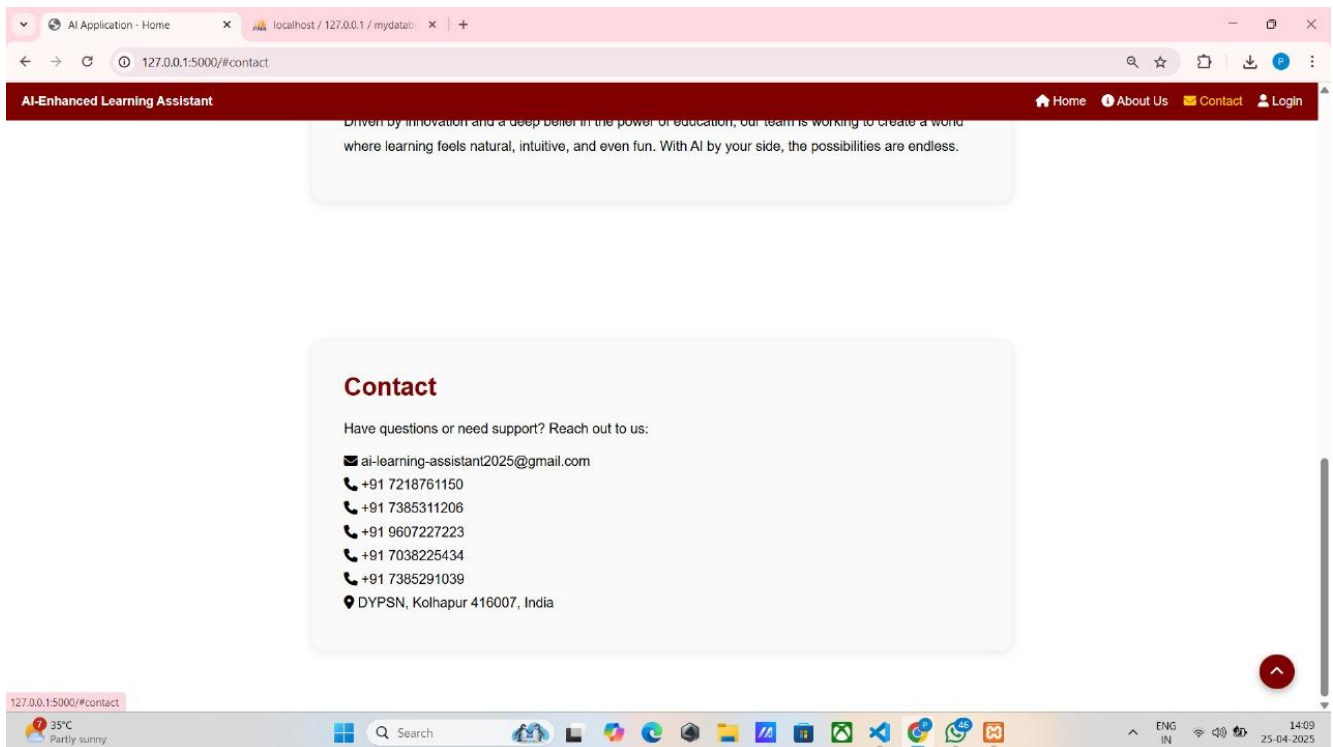
Home Page:



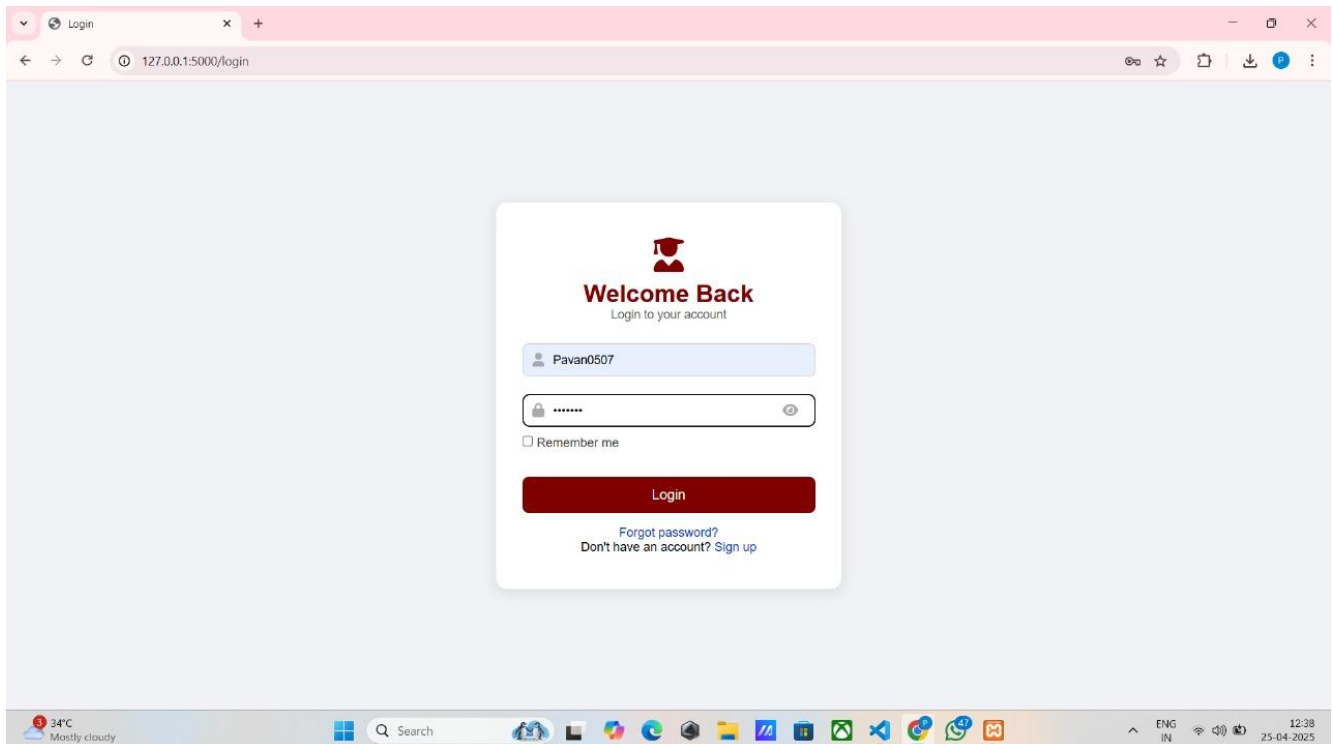
About us Page:



Contact:



Login:



Sign up:

+ Sign Up

Pavan

Jadhav

jpavan...@gmail.com

jpavan05

Sign Up

[Already have an account? Login](#)

Reset Password:

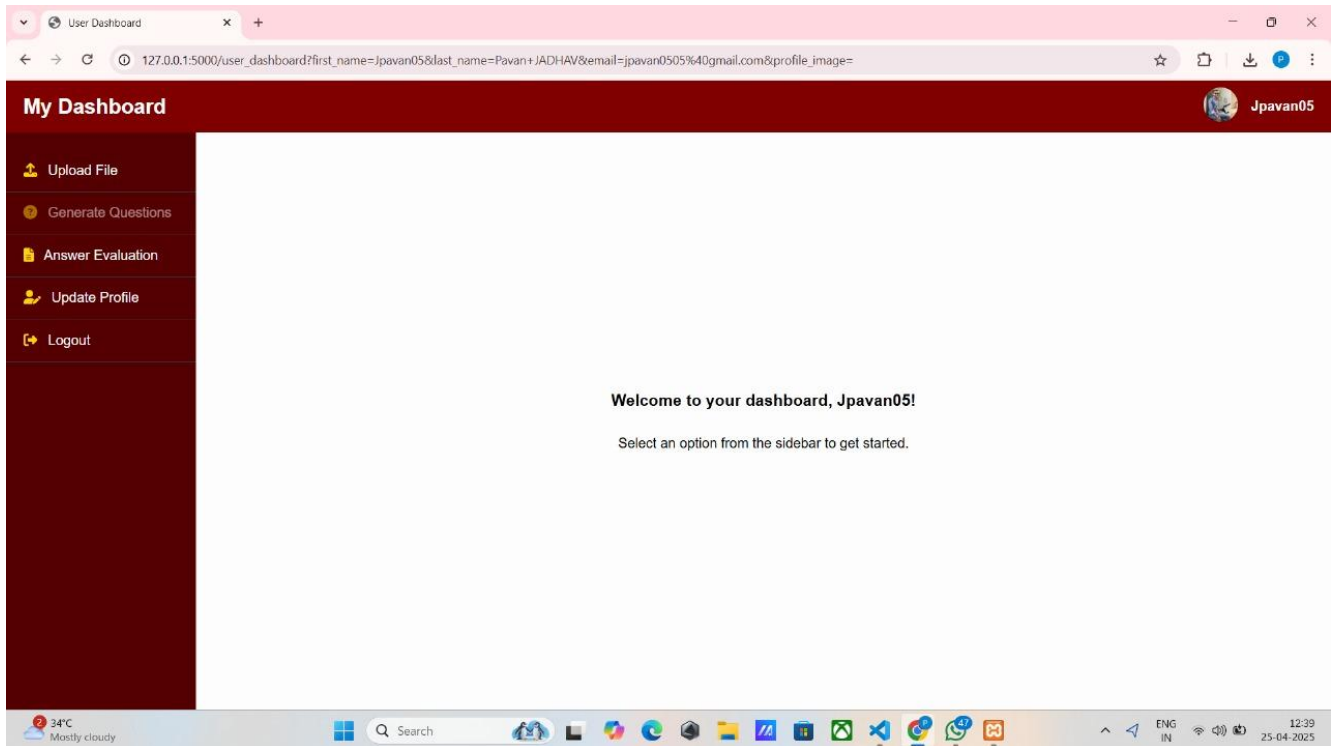
Forgot Password

jpavan...@gmail.com

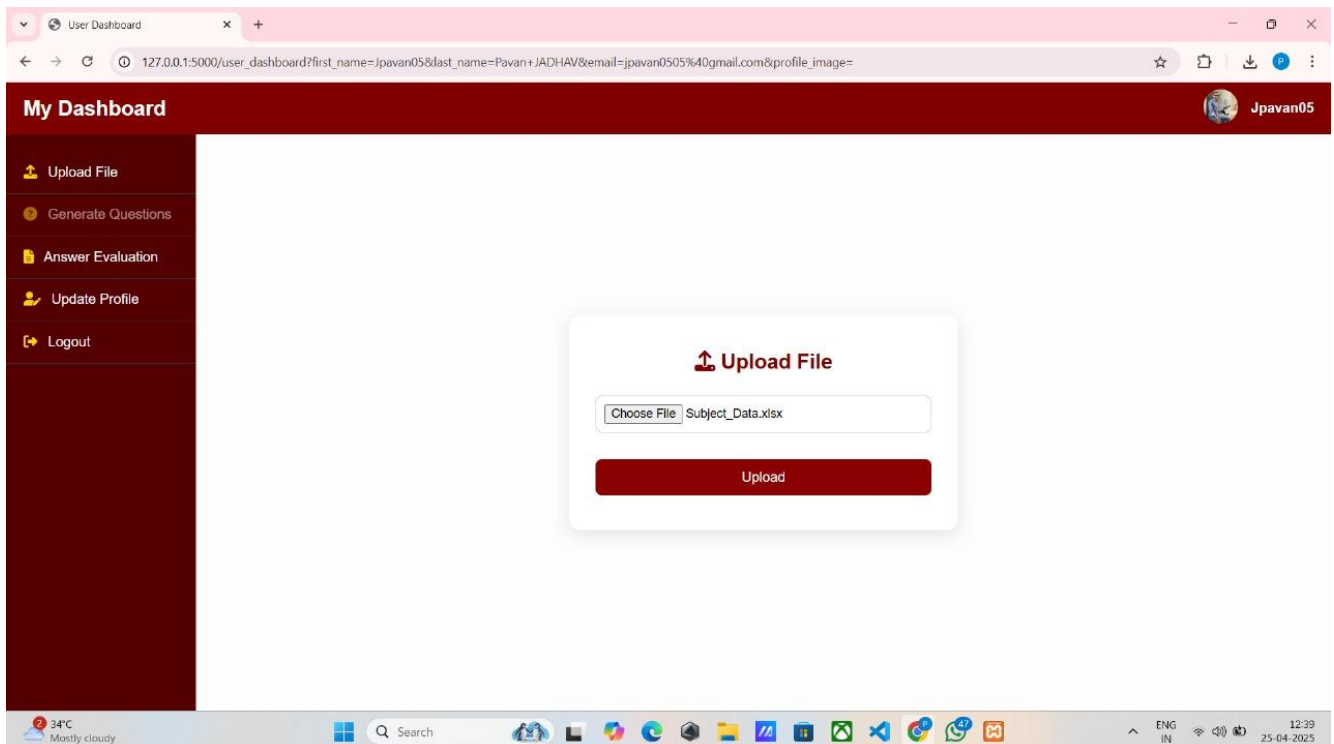
Send Reset Link

[Back to Login](#)

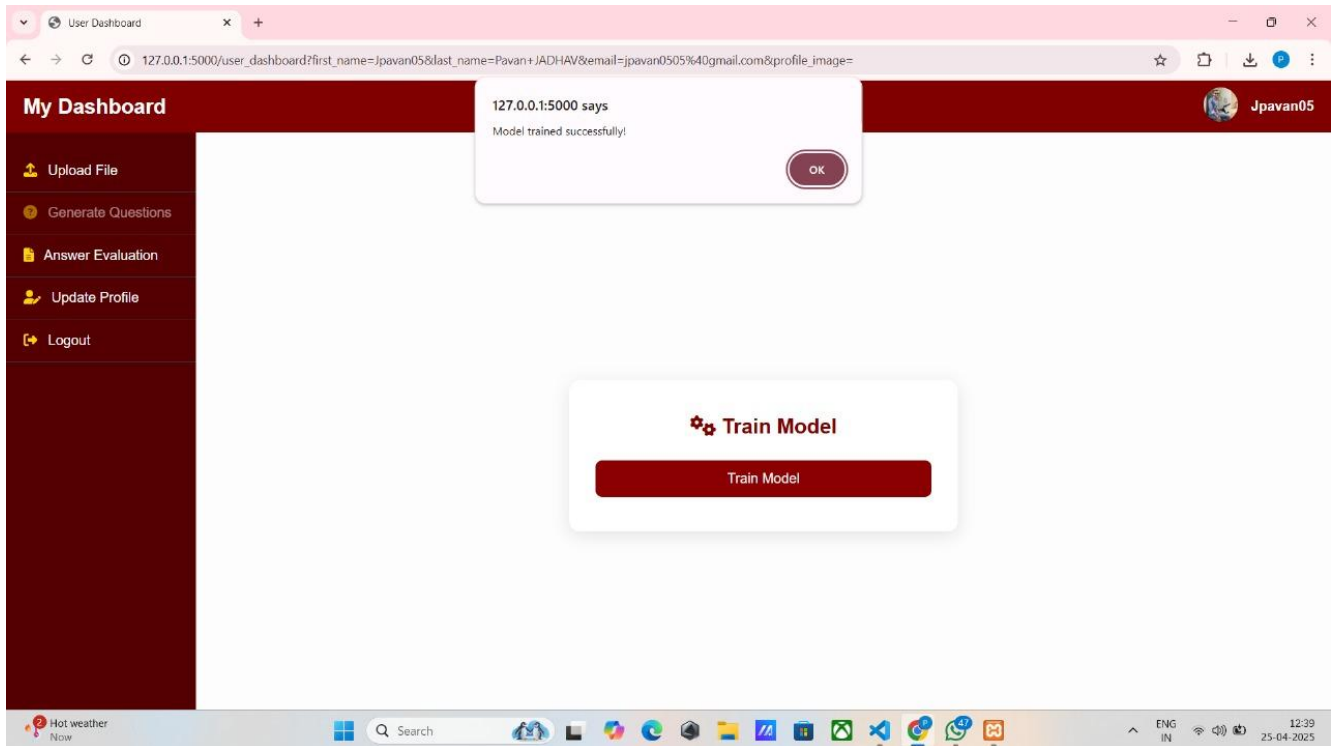
Dashboard:



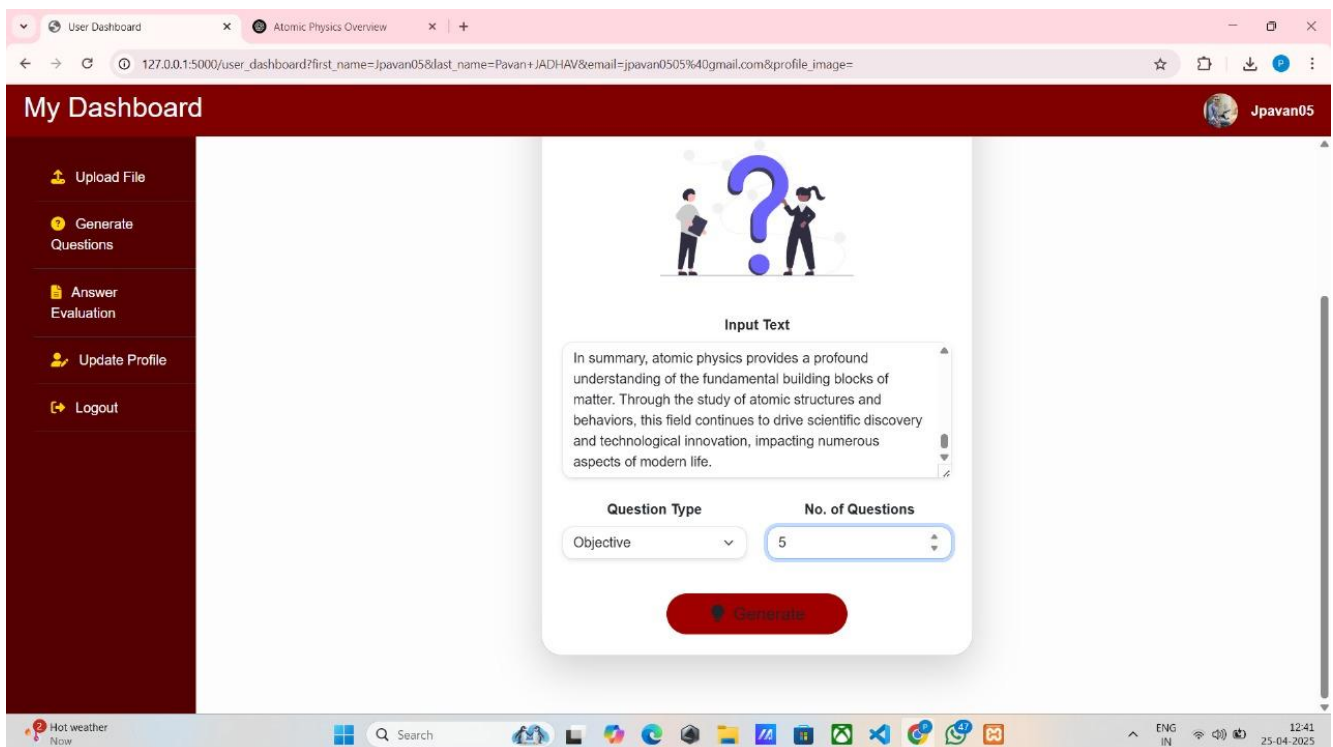
Upload Data:



Train Model:



Question Generation (Objective):



Objective Question Result:

The screenshot shows a web browser window with the URL `127.0.0.1:5000/question_generation`. The page features a header with a logo and a main content area titled "Generated Q&A". Below the title are buttons for "Copy", "CSV", "Excel", "PDF", and "Print", along with a search bar. The content is displayed in a table with two columns: "Question" and "Answer".

Question	Answer
_____ in technology have further expanded the horizons of atomic physics.	Advancements
_____ philosophers like Democritus and Kanāda postulated the existence of indivisible particles, which they termed "atoms."	Ancient
_____ physics is a fundamental branch of physics that delves into the intricate structure of atoms, focusing primarily on the behavior and interactions of electrons surrounding the nucleus.	Atomic
_____’s wave equation, in particular, allowed for the calculation of electron probability distributions, leading to the concept of atomic orbitals.	Schrödinger
_____ also explores the interactions between atoms and electromagnetic radiation.	Atomic physics

At the bottom of the table, it says "Showing 1 to 5 of 5 entries". Navigation buttons "Previous", "1", and "Next" are visible.

Question Generation (Subjective):

The screenshot shows a web browser window with the URL `127.0.0.1:5000/user_dashboard?first_name=Jpavan05&last_name=Pavan+JADHAV&email=jpavan0505%40gmail.com&profile_image=`. The page has a dark red header with the title "My Dashboard" and a user profile icon labeled "Jpavan05". A sidebar on the left contains navigation links: "Upload File", "Generate Questions", "Answer Evaluation", "Update Profile", and "Logout".

The main content area features a large blue question mark icon and an "Input Text" box containing the following text:

In summary, atomic physics provides a profound understanding of the fundamental building blocks of matter. Through the study of atomic structures and behaviors, this field continues to drive scientific discovery and technological innovation, impacting numerous aspects of modern life.

Below the input text, there are two fields: "Question Type" (set to "Subjective") and "No. of Questions" (set to "5"). A red "Generate" button is located at the bottom of the form.

Subjective Question Result:

The screenshot shows a web browser window with the URL `127.0.0.1:5000/question_generation`. The page features a cartoon character holding a sign with a question mark. Below the character is a section titled "Generated Q&A" with buttons for "Copy", "CSV", "Excel", "PDF", and "Print". A search bar is also present. The table below shows two generated Q&A entries:

Question	Answer
Define] FOR EXAMPLE.	[5] For example, there are two types of tin oxide: one is a grey powder that is 88.1% tin and 11.9% oxygen, and the other is a white powder that is 78.7% tin and 21.3% oxygen.
Explain in detail NITROGEN DIOXIDE.	[8][9] As a final example: nitrous oxide is 63.3% nitrogen and 36.7% oxygen, nitric oxide is 44.05% nitrogen and 55.95% oxygen, and nitrogen dioxide is 29.5% nitrogen and 70.5% oxygen.

Showing 1 to 2 of 2 entries

Previous 1 Next

Answer Evaluation (Upload Data):

The screenshot shows a web browser window with the URL `127.0.0.1:5000/user_dashboard`. The page is titled "My Dashboard" and features a sidebar with navigation options: "Upload File", "Generate Questions", "Answer Evaluation", "Update Profile", and "Logout". The main content area displays a section titled "Upload Student Answer" with a "Choose File" button and a file named "evaluation_result (4).pdf". Below this is a red "Upload and Evaluate" button. A note at the bottom states: "Make sure your file is in the correct format for evaluation."

Answer Evaluation Result:

My Dashboard Jpavan05

- Upload File
- Generate Questions
- Answer Evaluation
- Update Profile
- Logout

Evaluation Result

Student Answer:

Evaluation Result
Student Answer:
Evaluation Result
Student Answer:

1. Define UNCERTAINTY PRINCIPLE?
Measurements of quantum systems show characteristics of both particles and waves (wave-particle duality), and there are limits to how accurately the value of a physical quantity can be predicted prior to its measurement, given a complete set of initial conditions (the uncertainty principle).
2. Explain in detail WAVE-PARTICLE DUALITY?
Measurements of quantum systems show characteristics of both particles and waves (wave-particle duality), and there are limits to how accurately the value of a physical quantity can be predicted prior to its measurement, given a complete set of initial conditions (the uncertainty principle).
3. What do you mean by QUANTUM CHEMISTRY?
[2]: 1.1 It is the foundation of all quantum physics, which includes quantum chemistry, quantum field theory, quantum technology, and quantum information science.
4. Write a short note on INFORMATION SCIENCE.
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Reference Answer:

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My Dashboard Jpavan05

- Upload File
- Generate Questions
- Answer Evaluation
- Update Profile
- Logout

Reference Answer:

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Score:
9.85/10

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Update Profile:

My Dashboard

Update Profile

Profile updated successfully

First Name:
Bajarang

Last Name:
Dhamanekar

Email:
bajarang2002@gmail.com

Profile Image:


Choose File No file chosen

Update Profile

34°C Mostly cloudy

Search

ENG IN 12:50 25-04-2025

CHAPTER 8
SYSTEM TESTING

8 System Testing

8.1. System Testing:

System Testing is a level of software testing where complete and integrated software is tested. The purpose of this test is to evaluate the system's compliance with the specified requirements. The process of testing an integrated system to verify that it meets specified requirements.

8.2. Test Results:

8.2.1. Test Cases for User:

Test Case ID	Test Case Objective	Test Data	Expected Result	Actual Result	Status
TC 01	Validate Login Functionality	Username and Password	Accepts valid credentials. Username: min 1 char, can include special characters. Password: includes letters, numbers, special chars. Redirect to dashboard.	Accepts valid credentials. Redirects to dashboard.	Pass
TC 02	Validate Register Functionality	Name, Email, Username, Password, Confirm Password	All fields required. Email format validated. Passwords must match. Redirect to login page.	Fields validated. Mismatch error shown. Redirect on correct input.	Pass
TC 03	Upload Dataset File	File input (CSV, JSON)	Accept valid files. Show error on invalid formats. Success message and preview on upload.	CSV/JSON accepted. Invalid files show error. Success message shown.	Pass
TC 04	Train Model on Uploaded Data	Uploaded dataset	Click "Train" -> show progress. On completion, show success.	Progress shown. Training completes with success message.	Pass
TC 05	Generate Questions	Trained model data	Generate questions from trained data. Display as list.	Questions generated and displayed correctly.	Pass

TC 06	Evaluate Answers	User answers	Compare with expected. Provide scores and feedback per question and total.	Evaluation successful. Feedback and scores displayed.	Pass
TC 07	Login with invalid credentials	Invalid username/password	Show error: "Invalid username or password."	Error shown. Access denied.	Pass
TC 08	Register with existing email/username	Existing user info	Show error: "User already exists."	Error shown.	Pass
TC 9	Register with missing required fields	Empty fields	Highlight missing fields. Show validation errors.	Errors displayed and fields highlighted.	Pass
TC 10	Upload unsupported file format	Upload .exe or .txt	Show error: "Unsupported file type."	Error displayed. Upload rejected.	Pass
TC 11	Upload large dataset file	File > 100MB	Upload accepted or delayed if within limits. Else, error shown.	Upload successful with acceptable delay.	Pass
TC 12	Train model without uploading data	No dataset	Warning: "Please upload a dataset first."	Warning shown.	Pass
TC 13	Interrupt training midway	Click "Cancel"	Cancel training. Show message: "Training cancelled."	Training stopped. Message shown.	Pass
TC 14	Generate questions with untrained model	No model	Alert: "Train the model before generating questions."	Alert shown correctly.	Pass
TC 15	Generate questions with poor dataset	Minimal/unstructured dataset	Show error or generate basic questions only.	Minimal questions shown with a message.	Pass
TC 16	Submit empty answer for evaluation	Blank answer	Mark unanswered or 0 score.	Marked as 0.	Pass
TC 17	Submit nonsensical answer	Random symbols or irrelevant input	Mark incorrect. Provide feedback if possible.	Marked incorrect. Basic feedback provided.	Pass
TC 18	System timeout during training	Network interruption/server timeout	Show error: "Training failed due to timeout. Please retry."	Timeout handled. Error shown.	Pass
TC 19	Handle simultaneous uploads	Concurrent users	Queue or handle uploads asynchronously.	Uploads managed successfully.	Pass
TC 20	Evaluate answer with multiple correct options	Multiple correct answers	Award full/partial marks based on correct inputs.	Marks awarded correctly.	Pass

CHAPTER 9
APPLICATIONS

9.1 Applications

- Minimizes Manual Effort
- Improves Work Efficiency & Saves Time
- Streamlines Academic Operations
- Cost-Effective and Customizable
- Built with Modern Web Technologies
- Accessible Anytime, Anywhere
- Supports Personalized Learning

CHAPTER 10
CONCLUSION

10.1 Conclusion:

The Automatic Question Generation and Evaluation System proposed is an end-to-end platform for automated question paper generation and intelligent answer evaluation that leverages natural language processing (NLP), machine learning, and web technologies to facilitate the automation of educational assessment. Among other features it provides user authentication, profile management, objective and subjective question generation, automatic similarity based evaluation, result generation in PDF format.

By including TF-IDF and cosine similarity in answer evaluation, the system provides reliable and scalable solution to compare subjective answers against a reference. Flask as backend is modular and easy to deploy and MySQL ensures safe storage of user's data. Also dynamic interfaces as well as file handling capabilities makes the application more user friendly and accessible.

This solution reduces manual labor, improves grading objectivity, and introduces an adaptive framework to be extended to the development of more sophisticated and intelligent evaluation models in the future. In general, the proposed solution contributes substantially to digital education and automated assessment systems, which fulfil the growing demand for intelligent education technologies

CHAPTER 11
FUTURE WORK

11.1 Future Work

- Multimodal Question Generation

Incorporating images, graphs, or tables into the question generation pipeline can make the system capable of producing questions for visually rich content, particularly useful in subjects like biology, geography, or physics.

- Plagiarism and Bias Detection

Future systems should incorporate mechanisms to detect and prevent question duplication or unintentional bias in generated questions.

- Feedback Loop from Users

Collecting feedback from students and educators can help fine-tune the model and ensure that the generated content aligns with curriculum standards and learning goals

The feature scope includes enhancing the system with multimodal capabilities for generating questions from images, graphs, and tables. It also aims to improve content quality through plagiarism detection, bias reduction, and user feedback integration.

11.2 References

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6. (In Press) Choudhury, M., & Bhattacharyya, P. Automatic Question Generation and Answering Using Natural Language Techniques.
Explains ways of providing insightful and relevant questions from text data.